Technical Report: Predicting Loyalty Points for Turtle Games

Business Objective: Predict loyalty points accumulation and inform business decisions to enhance marketing efforts and the loyalty programme.

Summary

Turtle Games aims to improve its overall sales performance by leveraging customer data and optimising its loyalty program. This report utilises a various model to analyse customer data and answer the following questions:

- Are loyalty points being calculated correctly?
- Are customers understanding how best to accumulate the points?
- Are the marketing teams strategically targeting the groups?

The Data, Cleaning and Exploration

The analysis was conducted using the Pythons programming language for deep analysis, as well as R programming language, chosen for its compatibility with Turtle Games' existing workflows. A dataset, "turtle_reviews.csv," was cleaned and pre-processed. The dataset contained information on customer demographics, purchase habits, and loyalty points.

In Python this involved:

- Checking the meta data of the imported csv. Naming:
    - clean_reviews; to make a linear regression model in python
    - df2; to build decision trees
    - df3; to use k-clusters model to identify the groups
    - reviews_data; was made to dive deeper into reviews using NLP
- View column names, and types to understand what is needed for each step
- Removing irrelevant columns (language and platform, there is zero variance) to create a DataFrame based on numeric numbers.
- Renaming columns for clarity (remuneration..k.. to annual_income_k, spending_score..1.100. to spending_score).
- Stored clean version for future analysis

In my explorations and understanding of the business objectives, I categorise the variables:

- **Predictors**: Gender, age, annual_income_k, spending_score, education.
- **Target**: Loyalty points.
- **Others:** Product, reviews and summary.

The bulk of the analysis was conducted in a jupyter notebook, to easily document and visualise the data set.

Loyalty Points Analysis

I started off splitting the data in test and train sets, using Ordinary Least Squares function to find the best fitting straight line, and a summary to get the results. In this process I found that vs loyalty points, the spending score had the most significant relationship, due to an R-squared value of 0.452, (45% accuracy). The next closes were income

Key Insights: Loyalty points are right-skewed, suggesting that most customers accumulate lower points, with a few accumulating very high points.

There's a positive correlation between loyalty points, annual_income_k, and spending_score, indicating that customers with higher income and spending tend to earn more loyalty point.

Decision Tree Prediction

Using decision tree for better understanding of the data structure. Using the clean data, I transformed categorical columns using pandas-get_dummies, for numerical values to create a decision tree regressor model. Using an accuracy table and checking importance features, I decided a depth of 4, and both annual_income_k and spending_score are significant features, agreeing with previous analyse. Further pruning the tree, I found the most optimal model giving me an accuracy of 94%. The tree output, allows us to see what customer with a certain range of spend score and income split the groups of points. The top level shows that spending_score is the strongest factor in predicting loyalty points.

Key Insights: The decision tree model visually represents the decision rules for predicting loyalty_points based on factors like annual_income_k, spending_score, and age. It highlights the most influential factors and their thresholds for different loyalty point levels.

K-Means Clustering

Next model was k-clustering. Knowing that customer score and annual_income_k are the most prevalent factors in the prediction of loyalty points; these were the focus of this model. I compared them on a pairplot to understand their relationship., and then used the elbow method and silhouette methods to decide on the number of cluster that should be analysed, five being the outcome. Looking at the results, the five clusters appeared to be well balanced. I did some analysis to see whether there was a correlation between the k number and unique values in other variables, there was no conclusive outcome. From this I have identified futrher a groups that should be prioritised according to the position on the Final K-Means plot.

Key Insights: Customers segment into distinct clusters based on their annual_income_k and spending_score. These clusters reveal different customer groups with varying levels of loyalty points and spending habits. This segmentation allows for targeted marketing campaigns tailored to each cluster's characteristics.

NLP (Sentiment Analysis)

Using the clean csv, I used NLP libraries to find patterns in the words used. To allow the libraries to work I changed the text to lowercase and removed special characters. After removing stopwords, plotting a word cloud and bar-plot to visualise the most common words. This overview gives us a good inside into what kind of what the overall thoughts of the customers. I also plotted sentiment and polarity distributions to find that the overall view appears to be neutral to positive. Polarity for both reviews and summary are heavily focused in the at 0, with slight weight towards positive. Sentiment scores skewed towards the right, indicating generally positive tones. Summary indicated that comment lack strong emotion, largely being at 0, but heavily skewed to the right, showing tendency to be positive. Conclusion is that the overall view is positive.

Key Insights: Analysis of customer reviews reveals frequent use of words like "game," "play," "fun," and "time," highlighting what matters to customers. Positive reviews emphasise enjoyment, quality, and value, while negative reviews focus on difficulty, glitches, and price, providing feedback on product strengths and weaknesses.

Linear Regression

Multiple Linear Regression (MLR) models were constructed in R to predict loyalty points based on various customer features. The models were evaluated for goodness of fit using statistical measures such as R-squared, adjusted R-squared, and p-values. Visualisations, including scatter plots and Q-Q plots, were employed to assess the model fit and interpret the relationships between variables.

Key Insights: The linear regression model (using log-transformed variables) quantifies the relationship between loyalty points and predictors age, annual_income_k and spending_score. It provides an equation to predict loyalty points for new customers based on their income and spending patterns.

Answering The Business Questions

Loyalty Point Prediction: The MLR model successfully predicted loyalty points based on annual_income_k and spending_score.

Model Confidence: The model showed reasonable accuracy, as indicated by the goodness of fit measures.

Marketing Focus: Marketing efforts should target customers with high annual_income_k and spending_score, as these are the strongest predictors of loyalty points accumulation.

Loyalty Program Improvement: Consider a tiered rewards system based on predicted loyalty points to motivate customer engagement.

Analysis Improvement: Include additional customer features (e.g., purchase history, gaming preferences) and explore alternative models to potentially enhance predictive power.

Recommendations

Loyalty Programme Improvements:

- **Target High-Spending Customers**: Spend more effort on segments showing high correlation with loyalty points.
- **Improve Engagement**: Focus on younger demographics or certain customer types.

Improved Analysis:

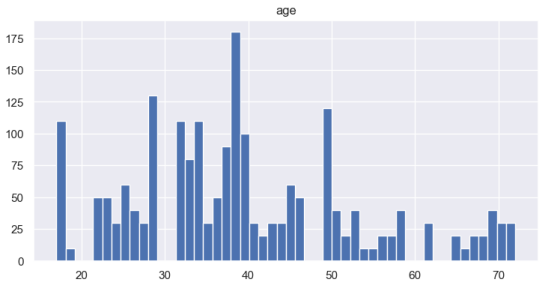
- Collect additional features (e.g. transaction frequency, product preferences).
- Explore advanced models like **Random Forest** or **Gradient Boosting** to improve predictions.

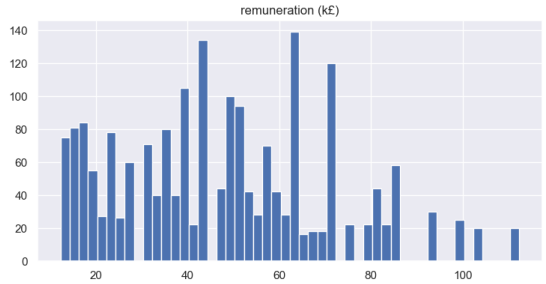
Supporting Visualisations and findings:

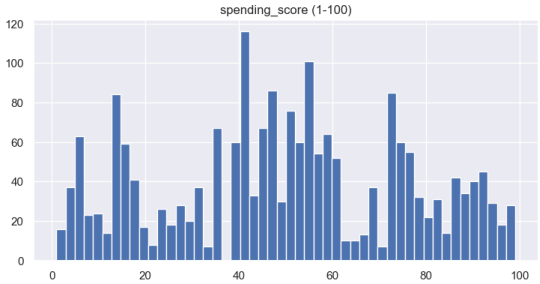
Loyalty Points Analysis:

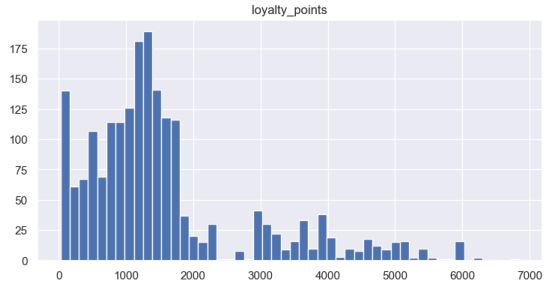
```
# To print just R-squared value of "spending_score".
print(f"Spending Score vs Loyalty Points Model R-squared: {results_spending.rsquared:.4f}")

# To print just R-squared value of "annual_income_k".
print(f"Annual Income (£k) vs Loyalty Points Model R-squared: {results_income.rsquared:.4f}")

# To print just R-squared value of "age".
print(f"Age  vs Loyalty Points Model R-squared: {results_age.rsquared:.4f}")
```

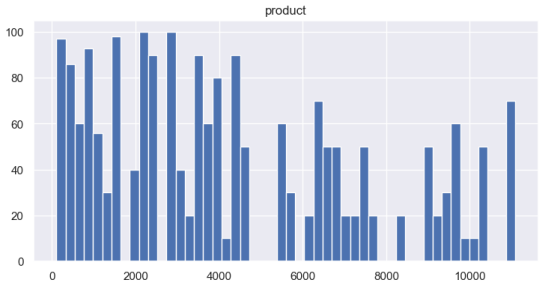
```
Spending Score vs Loyalty Points Model R-squared: 0.4520
Annual Income (£k) vs Loyalty Points Model R-squared: 0.3795
Age  vs Loyalty Points Model R-squared: 0.0018
```



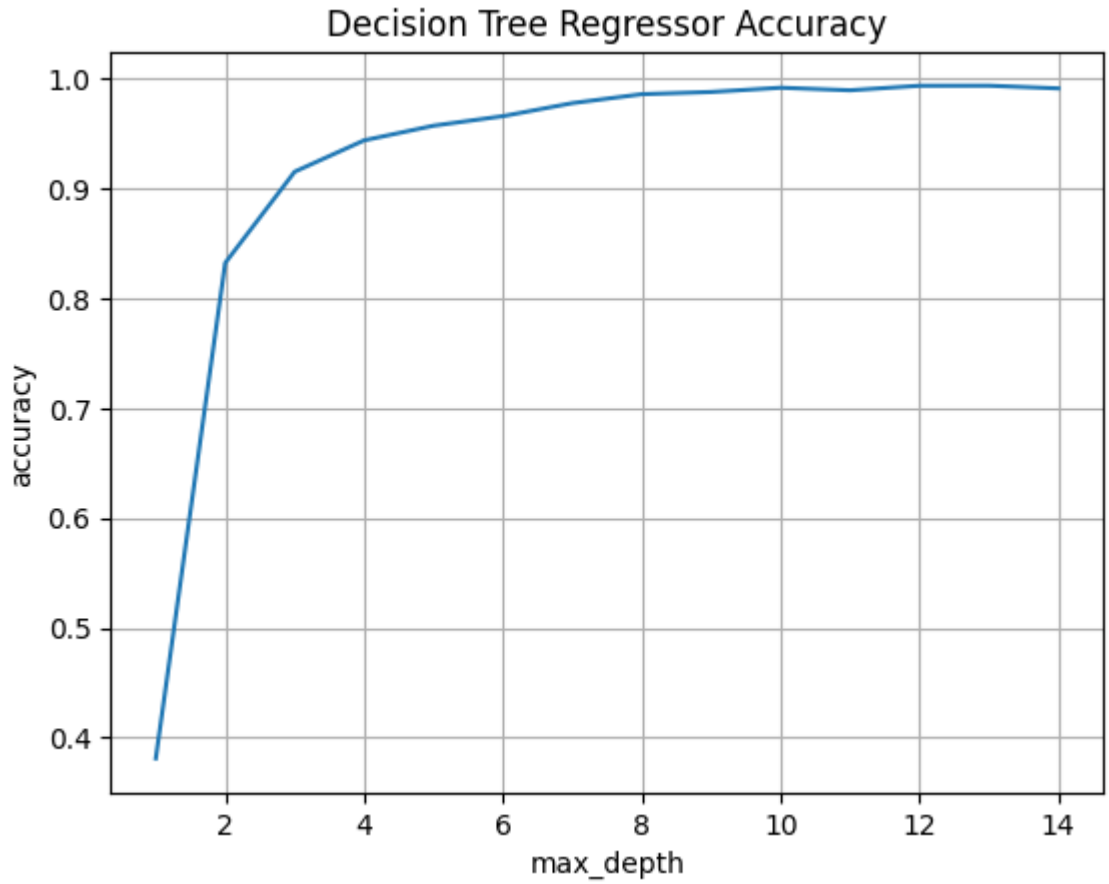








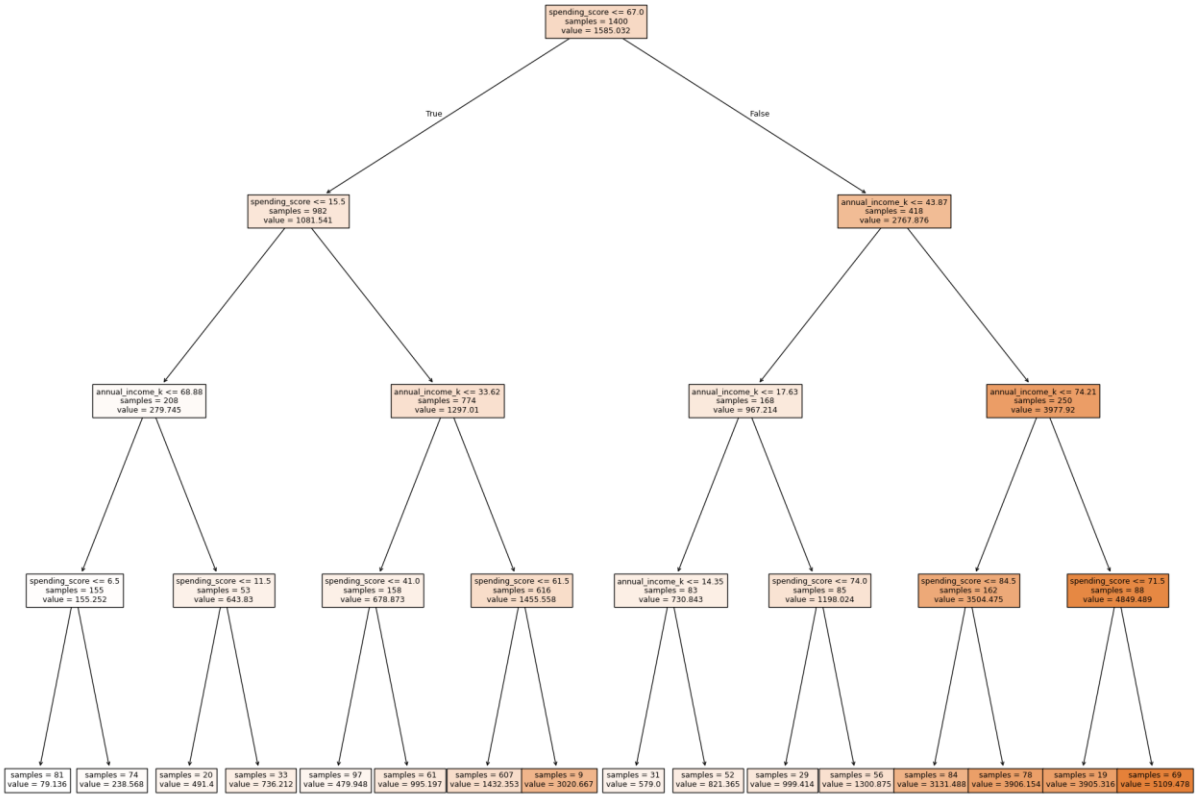
Decision Tree Prediction:



```python
# View the feature importance's, to see which features are most important.
importances = pd.DataFrame({
    'feature':X_train.columns,
    'importance':np.round(regressor.feature_importances_,3)
    })

importances = importances.sort_values(
    'importance',
    ascending=False
    )

importances[0:10]
```

	feature	importance
1	annual_income_k	0.498
2	spending_score	0.483
0	age	0.016
3	gender_Female	0.000
4	gender_Male	0.000
5	education_Basic	0.000
6	education_PhD	0.000
7	education_diploma	0.000
8	education_graduate	0.000
9	education_postgraduate	0.000

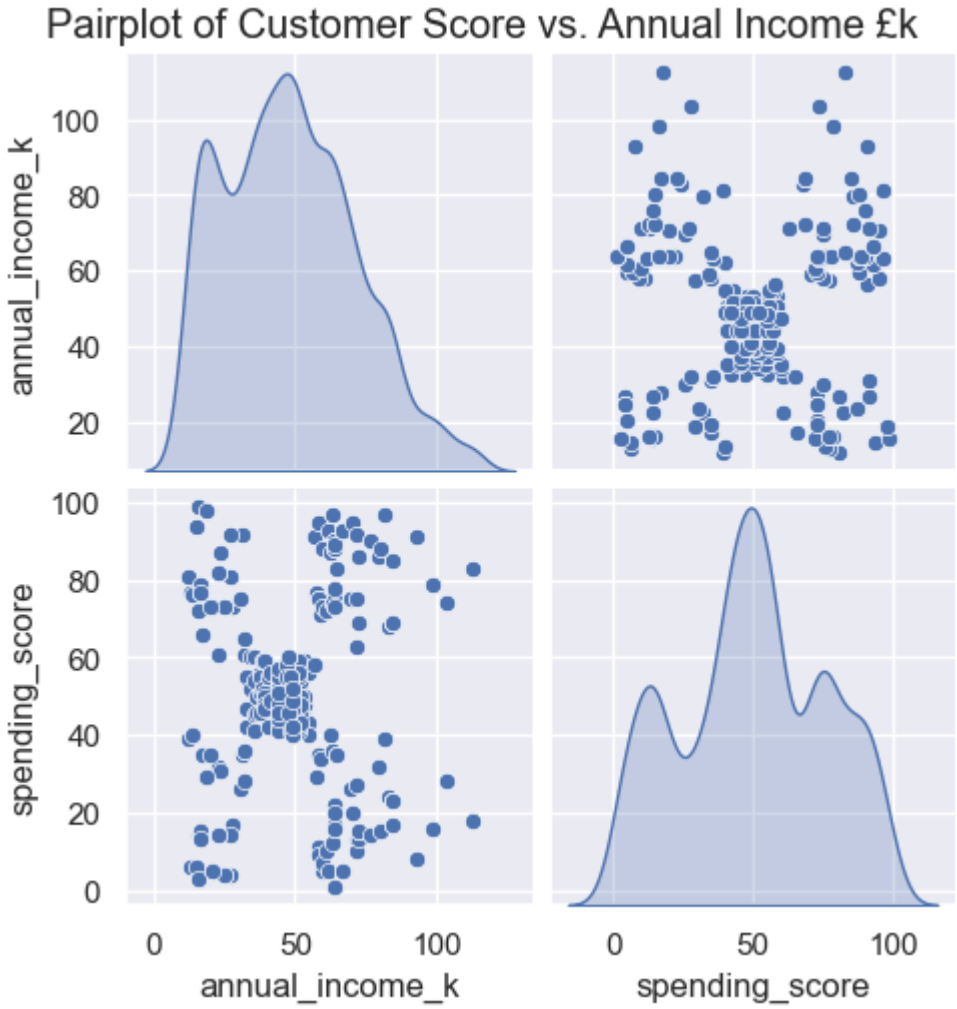
Pruned Decision Tree

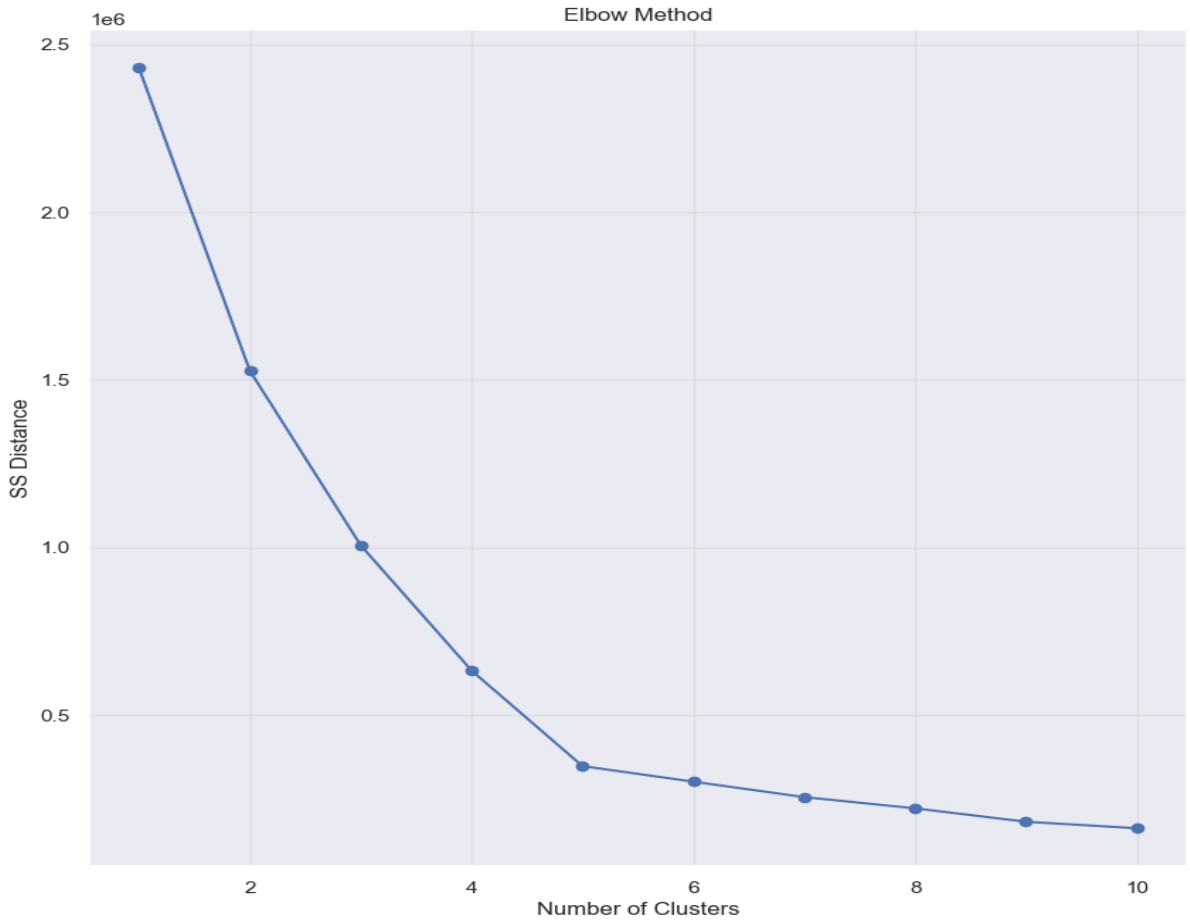


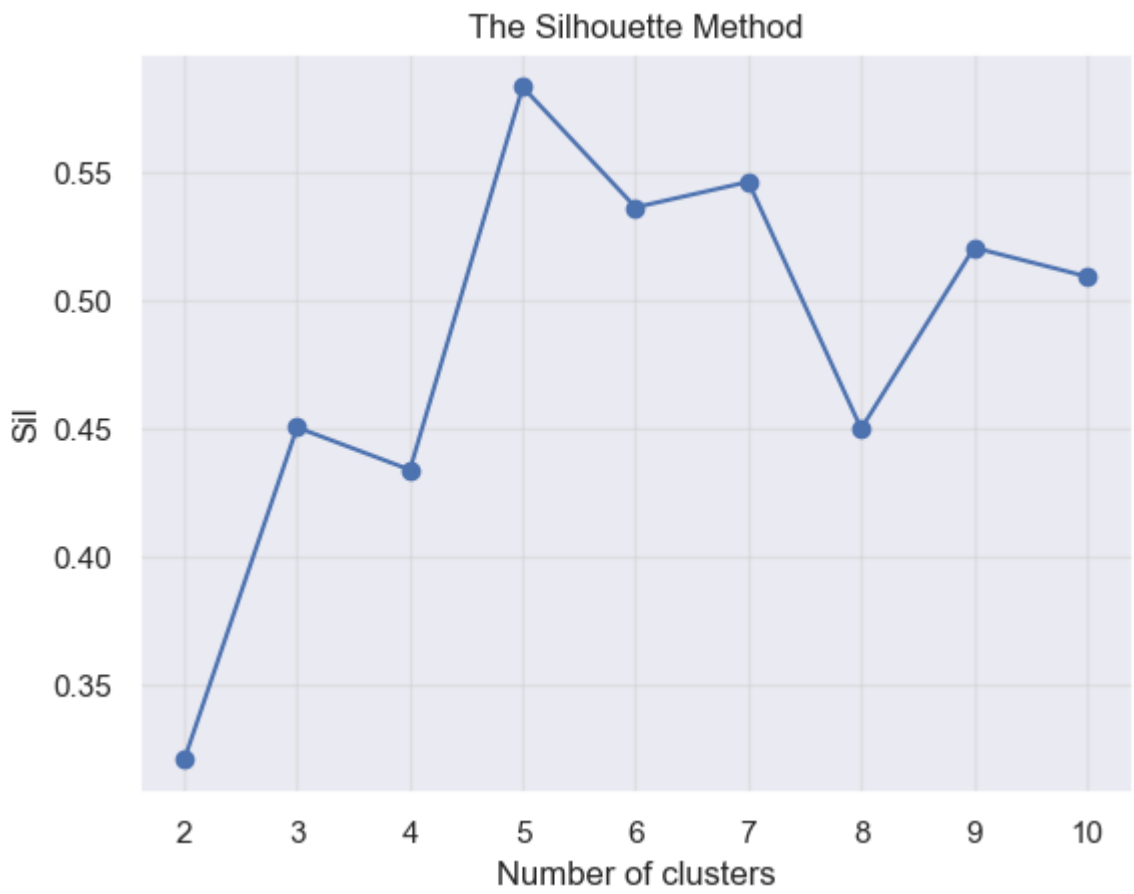
```python
# The final model.
f_regressor = DecisionTreeRegressor(
    random_state=42,
    max_depth=4,
    min_samples_split=5
    )

# Fit the model.
f_regressor.fit(X_train_new, y_train)

# Print the Regression models information.
print("FINAL DECISION TREE REGRESSOR MODEL")

# Predict the response for the data test.
y_predf = f_regressor.predict(X_train_new)

# Print additional information regarding the tree created.
print("Depth =",f_regressor.get_depth())
print("Leaves =",f_regressor.get_n_leaves())

# Specify to print the MAE and MSE (to evaluate the accuracy of the new model).
print(
    "\nMean Absolute Error: ", metrics.mean_absolute_error(y_train, y_predf)
    )
print(
    "Mean Squared Error: ", metrics.mean_squared_error(y_train, y_predf)
    )
# Calculate the RMSE.
print(
    "Root Mean Squared Error: ",
     math.sqrt(metrics.mean_squared_error(y_train, y_predf))
     )
print("R2 Score: ", r2_score(y_train, y_predf))
```

```
FINAL DECISION TREE REGRESSOR MODEL
Depth = 4
Leaves = 16

Mean Absolute Error:  211.21851310350354
Mean Squared Error:  89324.90757172268
Root Mean Squared Error:  298.87272804945366
R2 Score:  0.9460890153610675
```

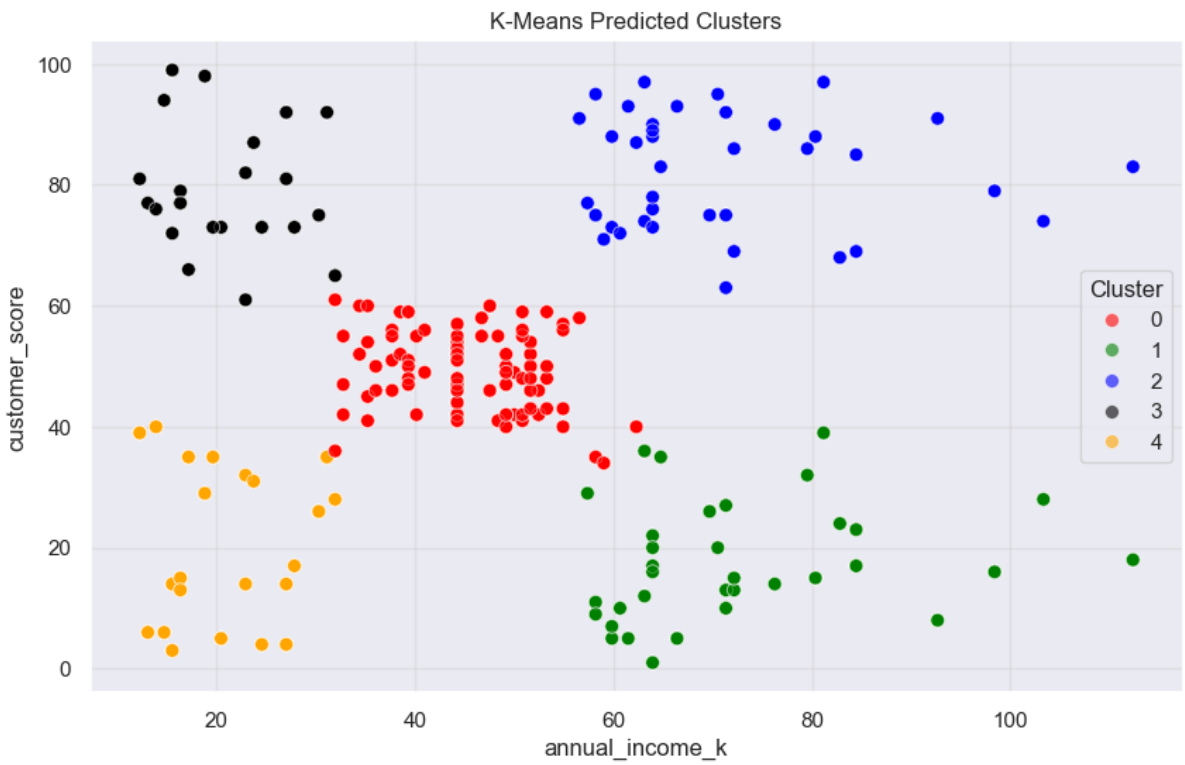
K-Means Clustering:




Elbow Method
1e6
2.5
2.0
1.5
1.0
0.5
SS Distance
2
4
6
8
10
Number of Clusters


The Silhouette Method
0.55
0.50
0.45
0.40
0.35
Sil
2
3
4
5
6
7
8
9
10
Number of clusters

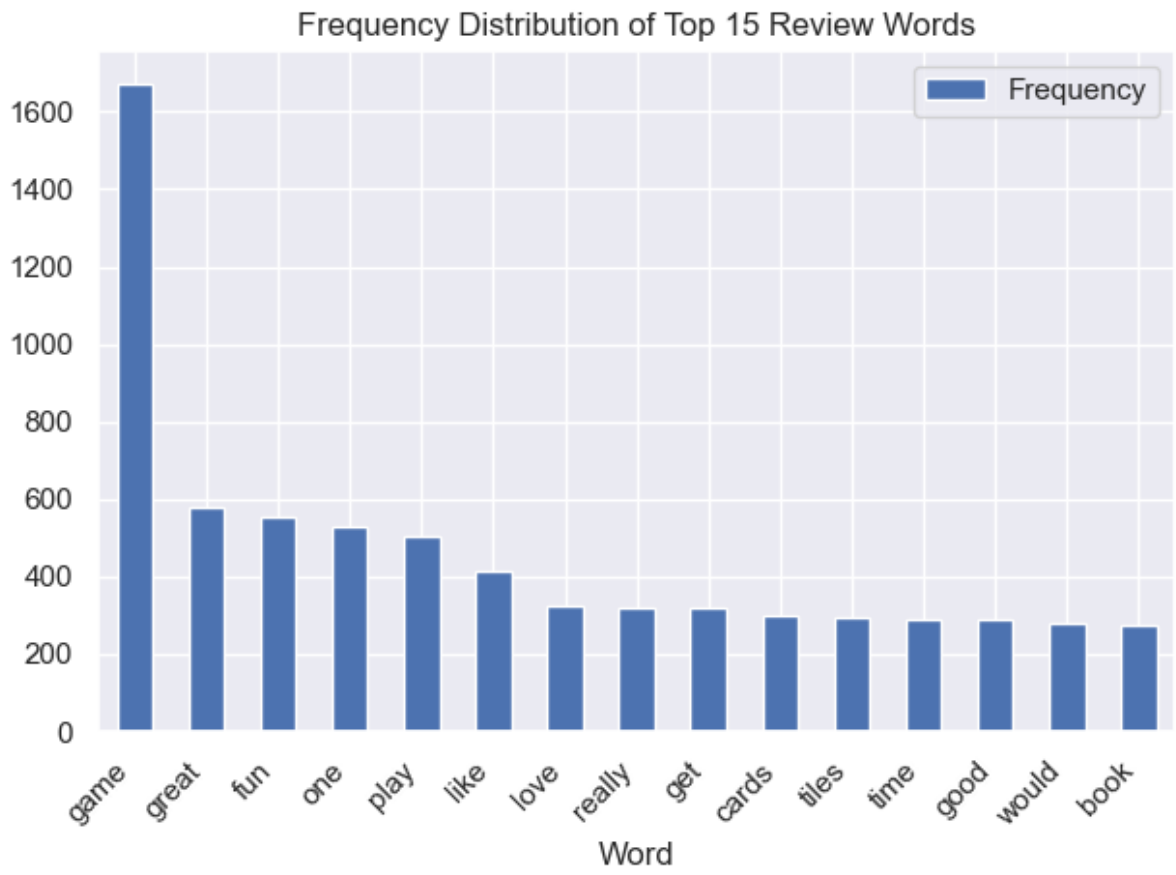
Pairplot of Customer Score vs. Annual Income £k with 5 Clusters
annual_income_k
spending_score
annual_income_k
spending_score
K-Means Predicted
0
1
2
3
4

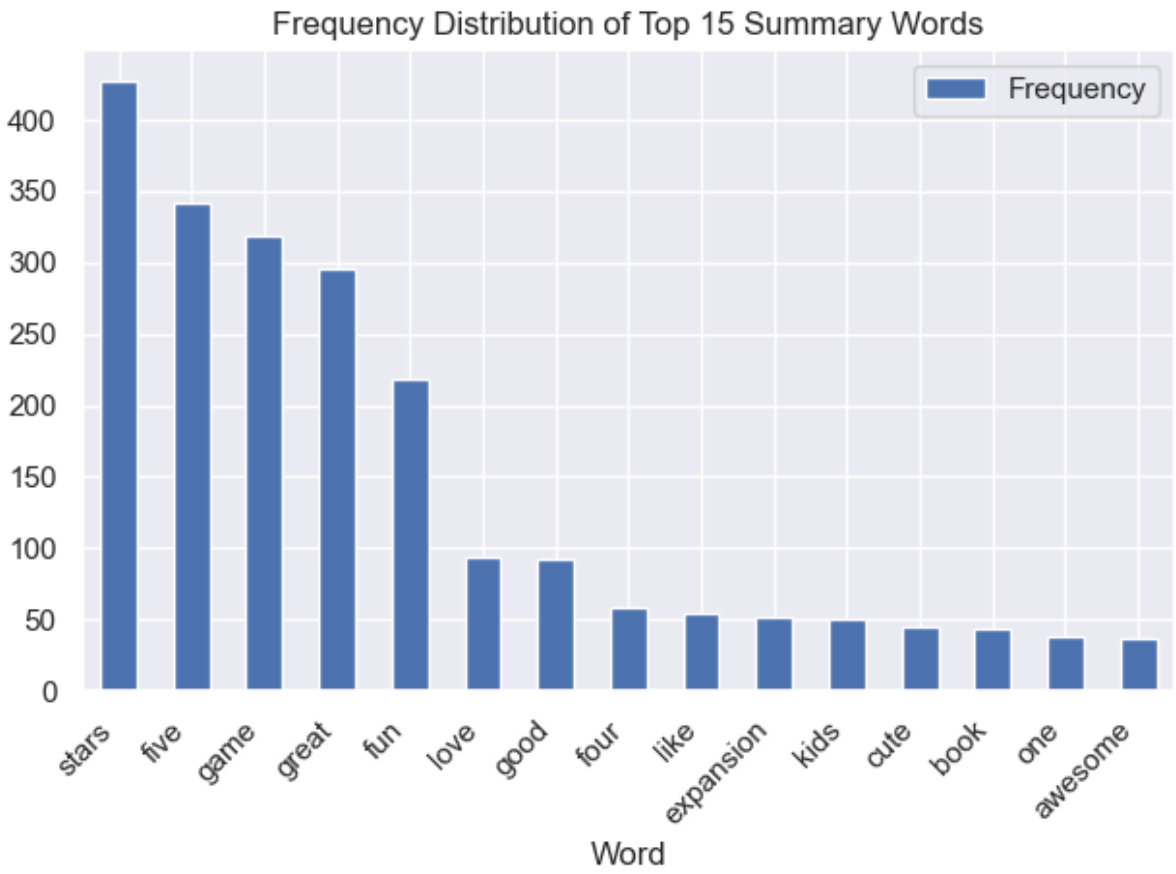

K-Means Predicted Clusters
customer_score
annual_income_k
Cluster
0
1
2
3
4

NLP (Sentiment Analysis):

Word Cloud of Reviews

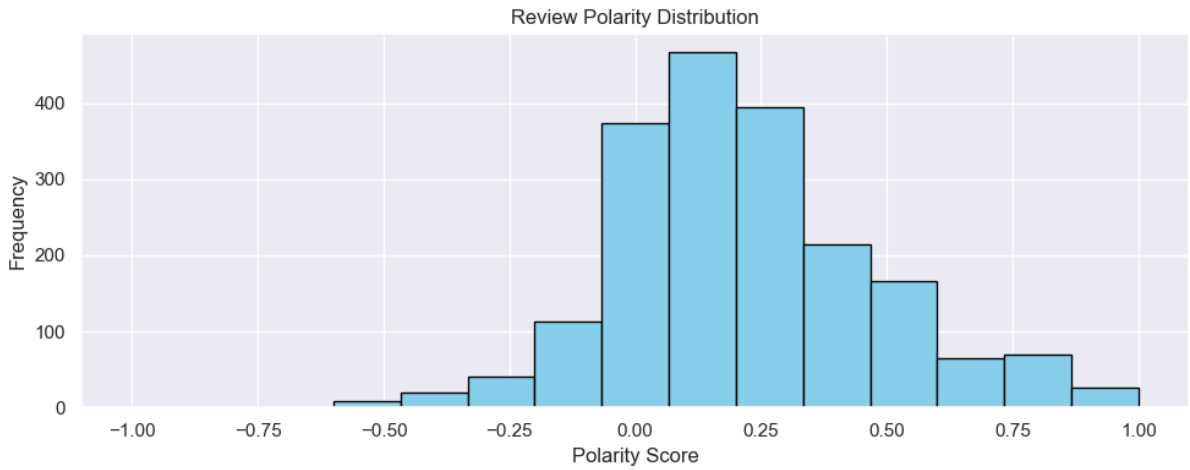
game
one
play
great
time
love
fun
tile
card
book
make
really
good
use
set
player
piece
well
new
expansion
little
product
still come
find
year old
thing
friend
shape
end
loved
look
see
nice
way
cute
much
help
awesome
played
first
monster
enough
playing
enjoy
board
family
rule
people
quality
dont
fun game
dungeon
strategy
color
better
bought
word
two
age
box
buy
got
miniature
board game
version
keep
give
lot
think
easy
gift
dd
many
made
toy
children
point
sticker
together
adventure
used
want
need
work
kid
back
different
great game
egg
cardboard
overall
instruction
ball whacks
lords waterdeep
letter
everything
pretty
daughter
looking
difficult
perfect
definitely
turn
bit
shape
friend
adult
love game
character
found
getting
animal
kit
puzzle
price
small
anger
question
hour
place
fit
always
child
hand
may
item
group
money
making
acquire
never
overall
cant
ive
actually
go
plastic
year
without
recommend
everyone
thought
feel
using
original
number
son
another
picture
others
take
something
going
already
hard
mini
best
start
every
know
three
building
quite
around
seem
might
play
game play
doesnt
quest
idea
able
far
youre
age
two
word
hard
bought
better
enjoy
playing
board
family
even
favorite
hand
first
may
item
always
child
monster
piece
time
well
great
good
use
really
make
tile
card
book
fun
love
different
perfect
definitely
price
puzzle
small
kit

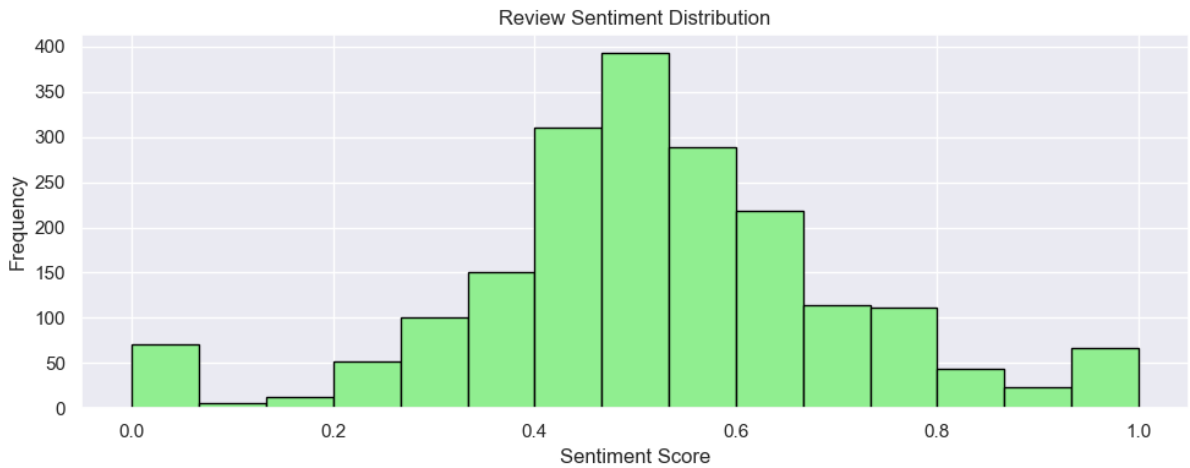

Frequency Distribution of Top 15 Review Words
Frequency
1600
1400
1200
1000
800
600
400
200
0
game
great
fun
one
play
like
love
really
get
cards
tiles
time
good
would
book
Word

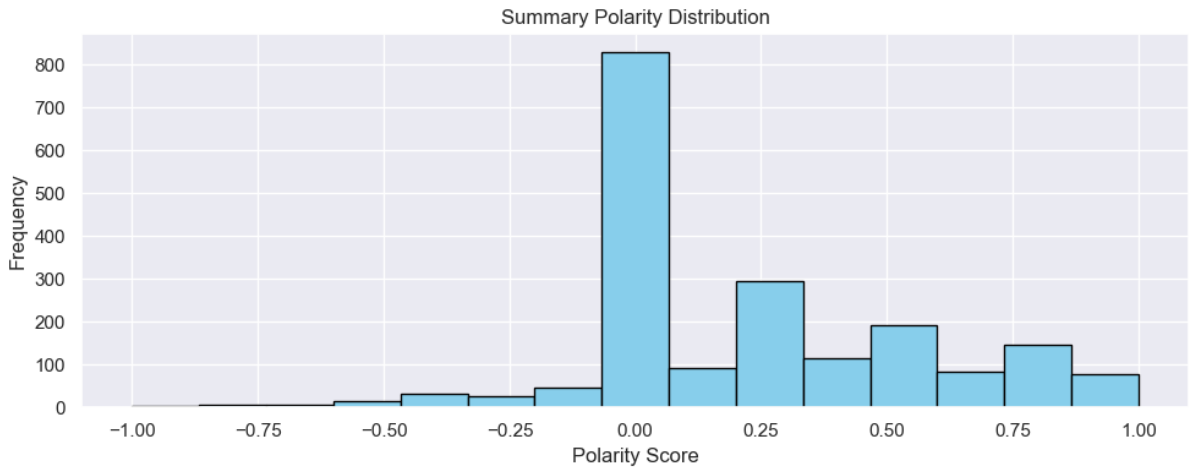

Frequency Distribution of Top 15 Summary Words
Frequency
400
350
300
250
200
150
100
50
0
stars
five
game
great
fun
love
good
four
like
expansion
kids
cute
book
one
awesome
Word

Word	Frequency in Reviews
game	1671
great	580
fun	552
one	530
play	502
like	414
love	323
really	319
get	319
cards	301
tiles	297
time	291
good	289
would	280
book	273

Word	Frequency in Summary
stars	427
five	342
game	319
great	295
fun	218
love	93
good	92
four	58
like	54
expansion	52
kids	50
cute	45
book	43
one	38
awesome	36

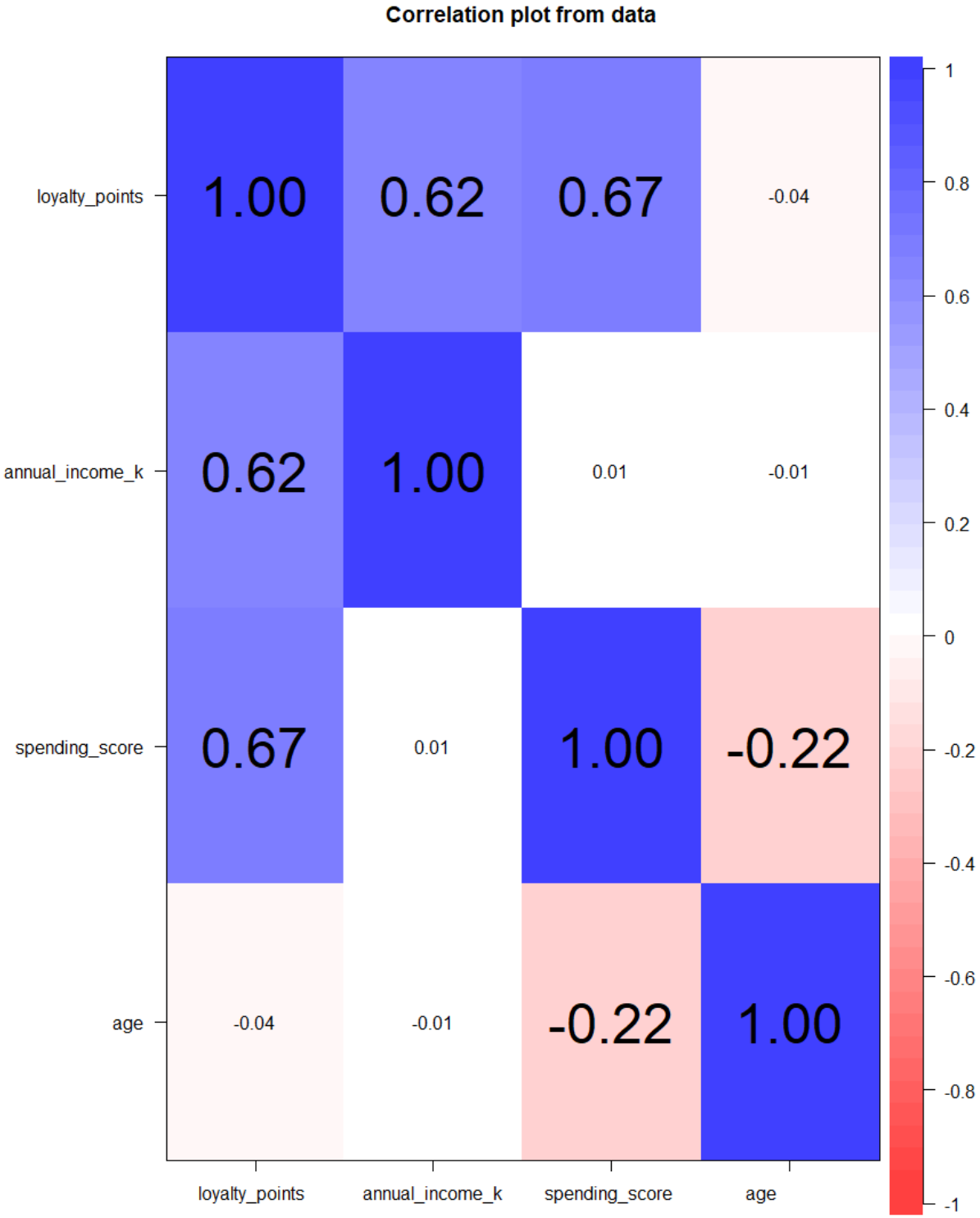

Review Polarity Distribution
Frequency
400
300
200
100
0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Polarity Score


Review Sentiment Distribution
Frequency
400
350
300
250
200
150
100
50
0
0.0
0.2
0.4
0.6
0.8
1.0
Sentiment Score


Summary Polarity Distribution
Frequency
800
700
600
500
400
300
200
100
0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Polarity Score

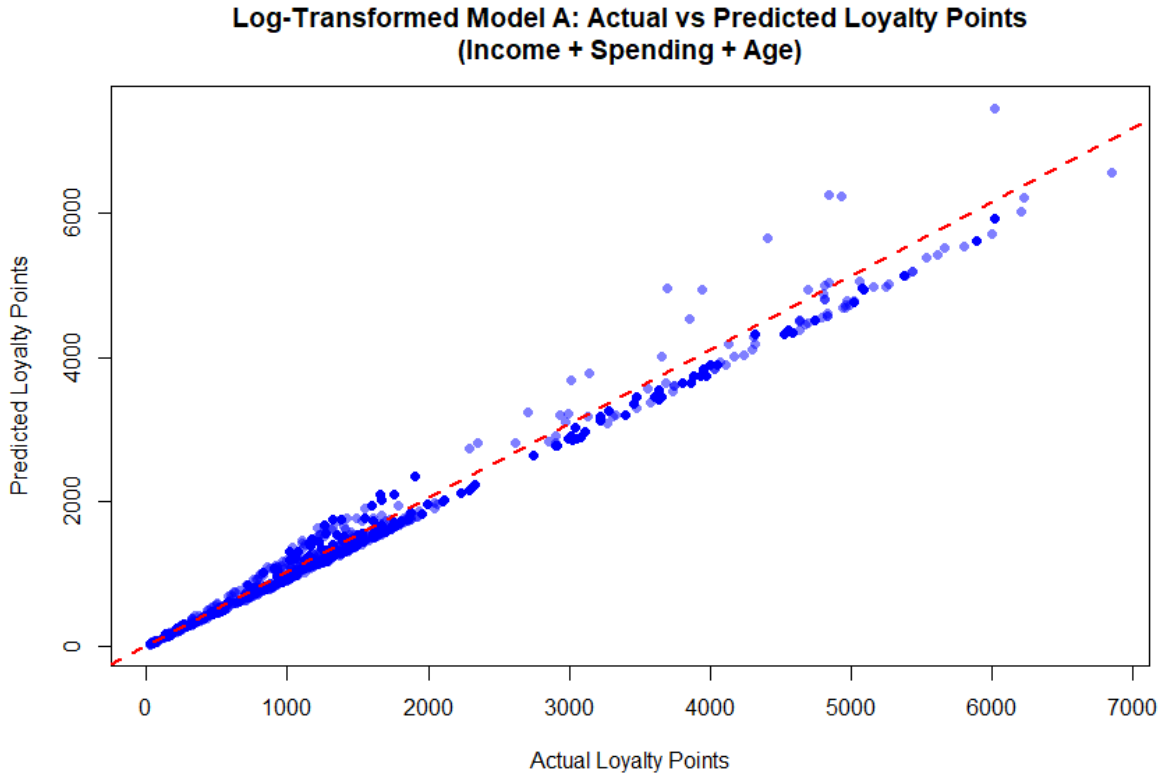
Summary Sentiment Distribution
Frequency
700
600
500
400
300
200
100
0
0.0
0.2
0.4
0.6
0.8
1.0
Sentiment Score

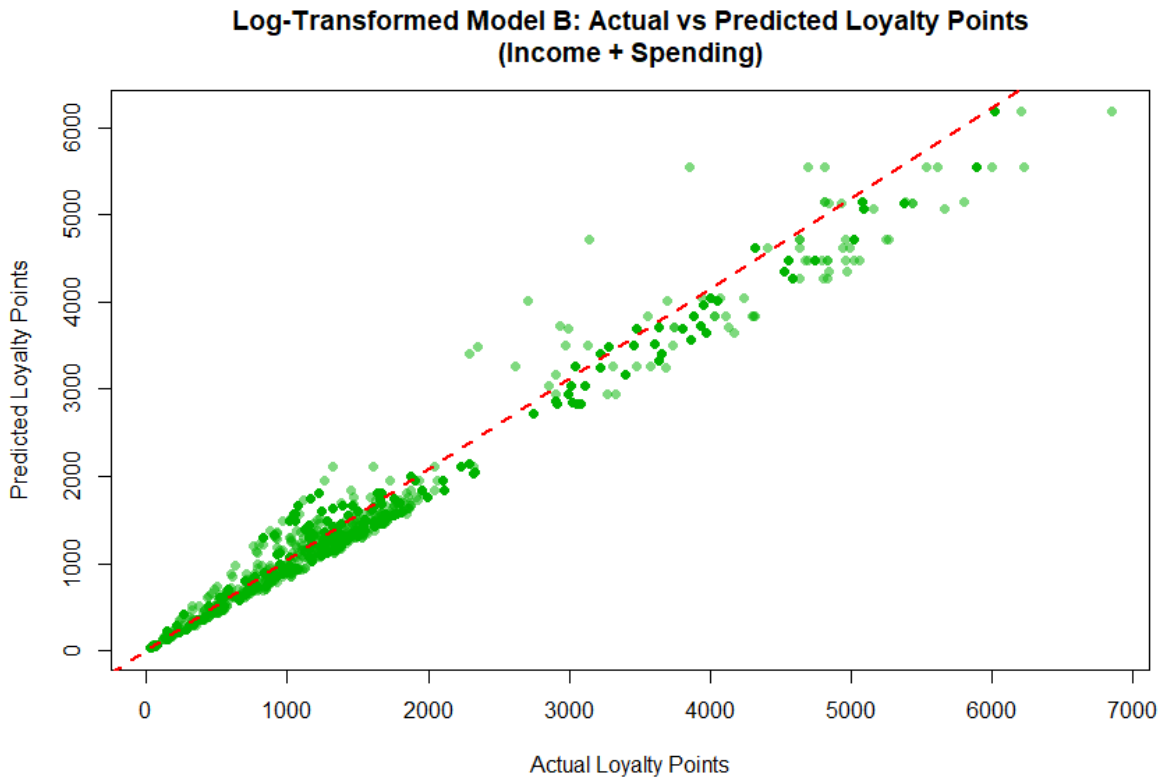
Linear Regression:

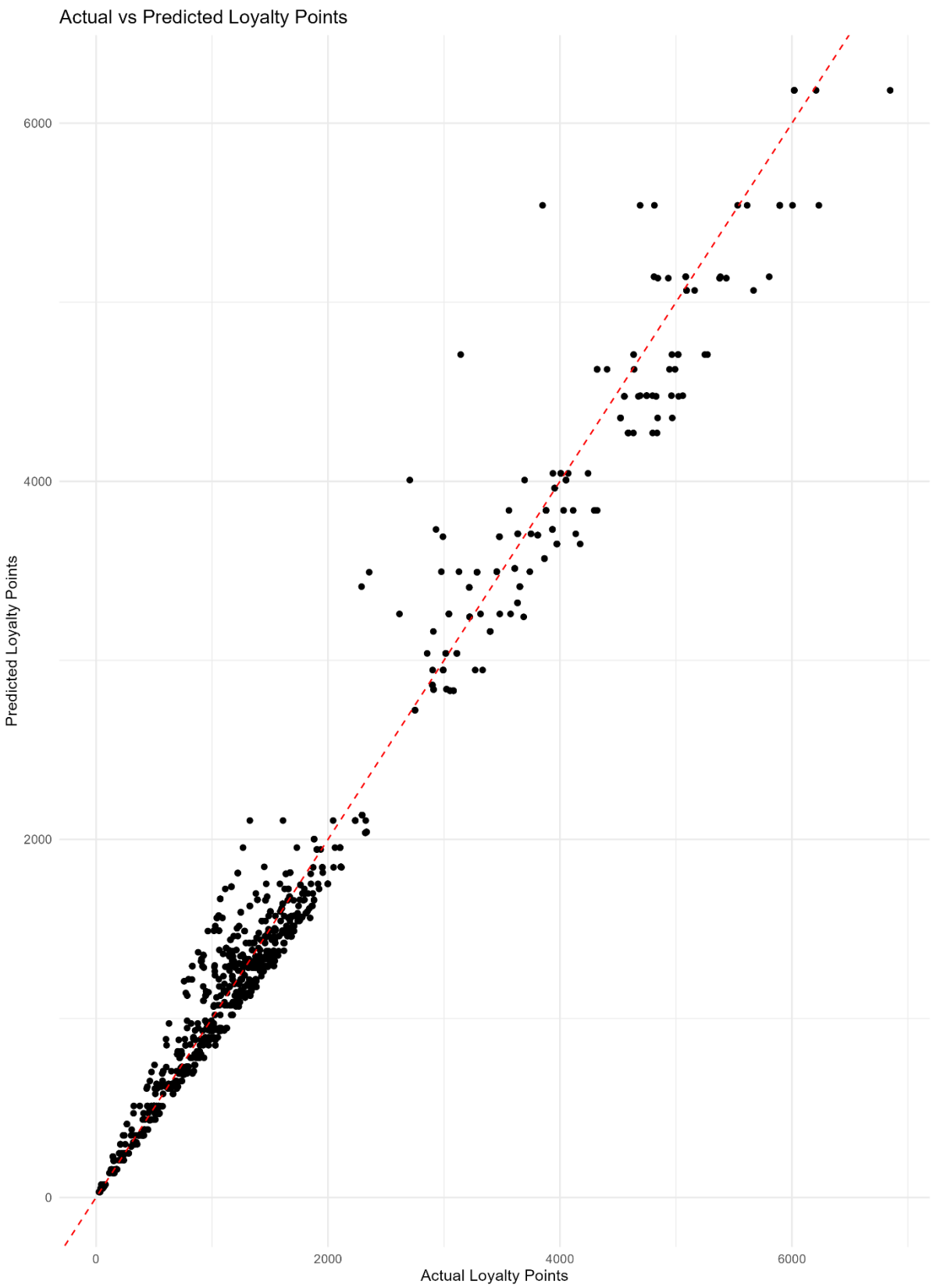


```r
# Log transform the better for better fitting of MLR
numeric_reviews <- numeric_reviews %>%
  mutate(
    log_loyalty_points = log(loyalty_points),
    log_annual_income = log(annual_income_k),
    log_spending_score = log(spending_score),
    log_age = log(age)
  )

# Plot the correlation.
plot(numeric_reviews$log_loyalty_points,
     numeric_reviews$log_annual_income + numeric_reviews$log_spending_score)


# Multiple Linear Regression Model
# Create the model using significant numeric predictors
# Model 1: All variables
model_a <- lm(log_loyalty_points ~ log_annual_income + log_spending_score + log_age,
              data = numeric_reviews)
# View the summary.
summary(model_a)
```

```
> summary(model_a)

Call:
lm(formula = log_loyalty_points ~ log_annual_income + log_spending_score +
    log_age, data = numeric_reviews)

Residuals:
     Min       1Q   Median       3Q      Max
-0.30612 -0.02391  0.03799  0.05932  0.10609

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         -1.639863   0.026914  -60.93   <2e-16 ***
log_annual_income    1.024546   0.003645  281.08   <2e-16 ***
log_spending_score   1.001138   0.002432  411.59   <2e-16 ***
log_age              0.307079   0.005820   52.76   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08942 on 1996 degrees of freedom
Multiple R-squared:  0.9923,    Adjusted R-squared:  0.9923
F-statistic: 8.611e+04 on 3 and 1996 DF,  p-value: < 2.2e-16
```


Log-Transformed Model A: Actual vs Predicted Loyalty Points
(Income + Spending + Age)
Predicted Loyalty Points
0
2000
4000
6000
0
1000
2000
3000
4000
5000
6000
7000
Actual Loyalty Points


Log-Transformed Model B: Actual vs Predicted Loyalty Points
(Income + Spending)
Predicted Loyalty Points
0
1000
2000
3000
4000
5000
6000
0
1000
2000
3000
4000
5000
6000
7000
Actual Loyalty Points


Actual vs Predicted Loyalty Points
Predicted Loyalty Points
Actual Loyalty Points
0
2000
4000
6000
0
2000
4000
6000